

# Lecture 18

— short proofs  
+ precise defs  
+ bijections

Proofs, Sets + Functions, Number Theory, Graph Theory, Counting

Goal: Count the number of objects satisfying a given property  
i.e. find the cardinality of  $S = \{x : P(x)\}$

ex 1: bit strings of length  $n$ ,  $S_n = \{(b_1, \dots, b_n) : b_i \in \{0, 1\}\}$

ex 2: bit strings of length 10 ending in 00 or starting with 1.

$$S = \{(b_1, \dots, b_{10}) : b_9 = b_{10} = 0 \text{ or } b_1 = 1\}$$

ex 3: Rankings of  $\{\text{Cal}, \text{Stanford}, \text{UCLA}, \text{USC}\} = U$   
 $S = \{(s_1, s_2, s_3, s_4) : s_i \in U, s_i \text{ distinct}\}$

ex 4: Simple graphs on  $V = \{1, \dots, n\}$ .  
 $S = \{G : G \text{ is a graph on } V\}$ .

# Principles of Counting

(Finite)

① If  $f: A \rightarrow B$  is a bijection, then  $|A| = |B|$ .

②  $|A \times B| = |A| \times |B|$ .

$(a_1, b_1) \dots (a_1, b_n)$

$\vdots$

$(a_m, b_1) \dots (a_m, b_n)$

③  $|A \cup B| = |A| + |B| - |A \cap B|$ .

Venn Diagram.  
"subtraction rule!"

Product Rule: Suppose each element of  $S$  is uniquely specified by a sequence of  $k$  choices  $c_1, c_2, \dots, c_k$

and # ways to make  $c_1 = n_1$

Given  $c_1$ , # "

$c_2 = n_2$

← doesn't depend on previous choices.

Given  $c_1, \dots, c_{k-1}$

$\vdots$

$c_k = n_k$

Then  $|S| = n_1 n_2 \dots n_k$  = # of ways to make  $(c_1, \dots, c_k)$ .

Proof:  
Sketch

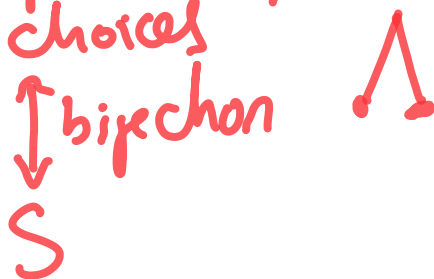
Choice tree.

vertices: partial choices

leaves: complete seq.  
of choices



↖ bijection  
paths from root

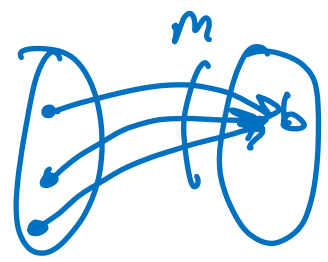


$(c_1, \dots, c_k)$

□

Division Rule: Defn:  $f: A \rightarrow B$  is m-to-1 if

$$\forall b \in B \quad |f^{-1}(\{b\})| = m.$$



Rule: If  $f: A \rightarrow B$  is m-to-1 then

$$|A| = m|B|, \text{ equivalently}$$

$$|B| = \frac{|A|}{m}.$$

ex 1: bit strings of length  $n$ ,  $S_n = \{ (b_1, \dots, b_n) : b_i \in \{0, 1\} \}$

$$\textcircled{2} : S_n = \{0, 1\} \times \dots \times \{0, 1\} \text{ } n \text{ times}$$

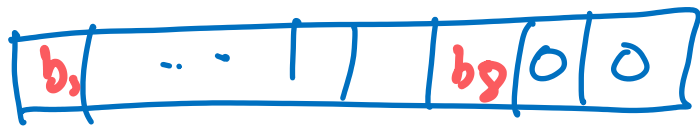
$$= \{0, 1\}^n$$

$$|S_n| = 2 \times 2 \dots \times 2 \text{ } n \text{ times} = 2^n$$

ex 2: bit strings of length 10 ending in 00 or starting with 1.

$$A \cup B = S = \{ (b_1, \dots, b_{10}) : \underline{b_9 = b_{10} = 0} \text{ or } b_1 = 1 \}$$

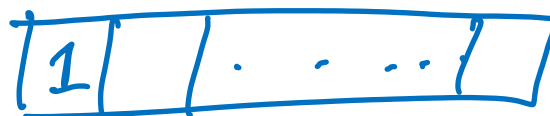
$$A = \{ (b_1, \dots, b_{10}) : b_9 = b_{10} = 0 \}$$



$$|A| = |S_A| = 2^8 \text{ bijection}$$

$$f(b_1, \dots, b_{10}) = (b_1, \dots, b_8)$$

$$B = \{ (b_1, \dots, b_{10}) : b_1 = 1 \}$$



$$|B| = |S_B| = 2^9$$

$$A \cap B$$

$$= \{ (b_1, \dots, b_{10}) : b_1 = 1 \wedge b_9 = b_{10} = 0 \}$$



$$|A \cap B| = |S_{A \cap B}| = 2^7$$

$$\textcircled{3} : |A \cup B| = 2^8 + 2^9 - 2^7$$

## II Product Rule

ex 3: Rankings of  $\{ \text{Cal}, \text{Stanford}, \text{UCLA}, \text{USC} \} = U$   
 $S = \{ (s_1, s_2, s_3, s_4) : s_i \in U, s_i \text{ distinct} \}$

idea: Decompose a <sup>full ranking</sup> big choice into a sequence of small choices.

① Process  
decompose

$C_1$ : choose  $s_1$

# ways  
4

$C_2$ : Given  $s_1$ , choose  $s_2 \neq s_1$ .

3

← independent of  $s_1$

$C_3$ : Given  $s_1, s_2$ , choose  $s_3$

2

$C_4$ : Given  $s_1, s_2, s_3$  choose  $s_4$

1

Upshot: **II** Total # of ways of making  $C_1, C_2, C_3, C_4$ :

$$= \begin{array}{l} \# \text{ ways of } C_1 \\ \times \# \quad \quad \quad C_2 \text{ given } C_1 \\ \times \quad \quad \quad \quad \quad \quad C_3 \text{ given } C_1, C_2 \\ \times \quad \quad \quad \quad \quad \quad \quad \quad C_4 \text{ given } C_1, C_2, C_3 \end{array}$$

$$= 4 \cdot 3 \cdot 2 \cdot 1 = \underline{\underline{24}}.$$

★ Each  $\underline{s} \in S$  <sup>onto</sup> <sup>1-1</sup> is uniquely specified by  
a sequence of choices.

$\{\text{choices}\} \xrightarrow{\text{bijection}} \{(s_1, s_2, s_3, s_4) : \text{ranking}\}$



$|S| = |\{\text{choices}\}| = 24$ . by principle ①.

Def: An ordering of  $n$  distinct objects is called a permutation.

Basic Count #1: The number of permutations of  $n$  distinct objects is  $n \times (n-1) \times \dots \times 1 = n!$ .  
< Proof: Product Rule.

ex 5:  $D =$  deck of cards      4 suits  
13 cards per suit.

$S = \{ (s_1, \dots, s_5) : s_i \in D, s_i \text{ distinct } \}$ .

① Process:

|                                         | ② #ways |
|-----------------------------------------|---------|
| $C_1$ : choose $S_1$                    | 52      |
| $C_2$ : Given $S_1$ , choose $S_2$      | 51      |
| $C_3$ : Given $S_1, S_2$ , choose $S_3$ | 50      |
| $C_4$ : " choose $S_4$                  | 49      |

C<sub>5</sub>:

" Choose 55 48

$$\text{Total ways} = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$$

$$\textcircled{\text{III}} \quad |S| = \frac{52 \cdot 51 \cdots 47 \cdot 46 \cdots 1}{47 \cdot 46 \cdots 1} = \frac{52!}{47!}$$

Product Rule.

Defn: An ordered sequence of  $r$  distinct objects chosen from  $n$  distinct objects is called an  $r$ -permutation.

Basic Count #2: The # of  $r$ -permutations of  $n$  distinct objects is

$$P(n, r) = \frac{n!}{(n-r)!}$$

<Proof: Product Rule>

### III Division Rule

ex6: Unordered sets of 5 distinct goods chosen from  $D$ .

$$S = \{ \{s_1, s_2, \dots, s_5\} : s_i \in D, \text{distinct} \}$$

Attempt: (I) Process A

$C_1$ : choose  $s_1$

$C_2$ : choose  $s_2 \neq s_1$  given  $s_1$

$\vdots$

Issue: Choices do not uniquely specify  $\{s_1, s_2, \dots, s_5\}$

Product rule does not apply.

eg.  $(1, 6, 11, 37, 19) \mapsto \{1, 6, 11, 37, 19\}$   
 $(6, 37, 11, 19, 1)$   
 $(37, 6, 19, 11, 1)$  Not 1-1

Observe that

$$f: \{ (s_1, \dots, s_5) : s_i \in D, \text{distinct} \}$$

$$\longrightarrow \{ \{s_1, \dots, s_5\} : s_i \in D, \text{distinct} \}$$

is m-to-1 for  $m=5!$  since there are  $5!$  orderings of each  $\{s_1, \dots, s_5\}$ .

By the division rule,  $|S| = \frac{52! / 47!}{5!} = \frac{P(52, 5)}{5!}$  ← ex 5.

Sneaky: Process for choosing  $(S_1, \dots, S_5)$ :

$C_1$ : Choose an unordered set  $\{S_1, \dots, S_5\}$   $|S|$  #ways

$C_2$ : Choose an ordering  $(S_1, \dots, S_5)$   $5!$

$\therefore$  Total #ways =  $\overset{\text{ex 5}}{P(52, 5)} = \overset{\text{Product}}{|S| \times 5!}$ .

$\therefore$   $|S| = \frac{P(52, 5)}{5!}$

Defn: An unordered set of  $r$  distinct objects chosen from  $n$  distinct objects is called an  $r$ -combination.

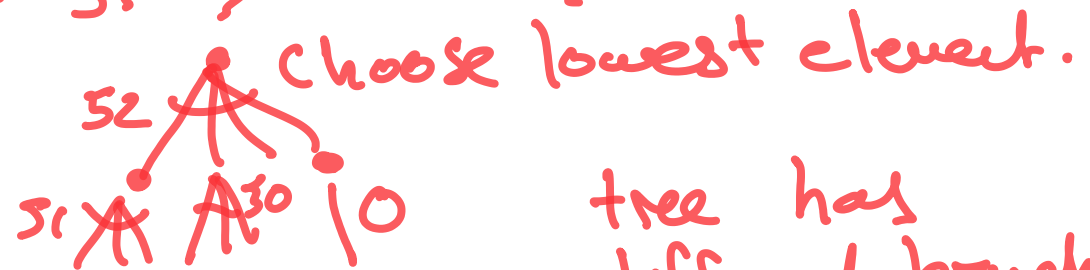
Basic Count #3: The number of  $r$ -combinations of  $n$  objects

derived  $\boxed{\binom{n}{r} \text{ is } \frac{n!}{r! (n-r)!}} = \frac{P(n,r)}{r!}$

also  $C(n,r)$ .

< Proof: Argument in ex6 >.

(Q) Tree for unordered choice?  $\xrightarrow{\text{bij}}$   $\{(s_1, \dots, s_5) : \text{sorted sequence}\}$



tree has different branching depending on choices.

Can't use product.

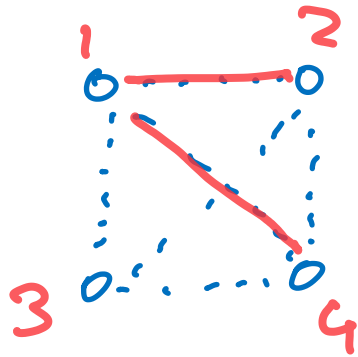
(Q) How many <sup>simple</sup>  $n$  graphs on  $\{1, \dots, n\}$ ?

$$G = (\{1, \dots, n\}, E) \quad \text{where}$$

$$E \subseteq \{T : T \subseteq \{1, \dots, n\}, |T|=2\}$$

$P$

ex:



$$n=4, \quad |P|=6.$$

$$n, \quad |P| = \binom{n}{2} \quad \text{in general.}$$

To uniquely specify  $G$ : Let  $P = \{P_1, \dots, P_{\binom{n}{2}}\}$ .

For each  $i = 1, \dots, \binom{n}{2}$

$C_i$ : decide if  $P_i$  is an edge. 2 <sup># ways</sup>

• By Product Rule: # ways =  $2 \times \dots \times 2 \binom{n}{2} \text{ times}$   
=  $2^{\binom{n}{2}}$ .

•  $S = \{ Q \text{ on } \{1 \dots n\} \}$  is in bijection

with  $\mathcal{P}(P)$ , which is in bijection

with bit strings on  $|P|$  bits,

So  $|S| = 2^{\binom{n}{2}}.$